

Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 7: A Tale of Three Intersecting Lines

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	7 — A Tale of Three Intersecting Lines
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. At this hardest tier every item chains several facts: the exterior angle equals the SUM of the two remote interiors (never the adjacent one); two sides must STRICTLY exceed the third (equal fails); 'acute' needs ALL three angles below 90° while a single 90° makes it right and any angle above 90° makes it obtuse; the largest angle always sits opposite the longest side; the angle between two interior bisectors of a triangle is $90^\circ + \text{half the third angle}$; an altitude is a perpendicular (not a side or median) and its foot can fall outside an obtuse triangle; and a count of integer third-side values must respect strict inequalities at both ends. Do not write on this paper — mark answers on your answer sheet.

1. In triangle ABC the angles are in the ratio 2 : 3 : 5. The bisector of the largest angle is drawn from its vertex to the opposite side. This bisector splits the largest angle into two equal parts; how large is each part?

- (A) 90° (B) 45°
(C) 30° (D) 36°

2. In triangle PQR, side QR is extended beyond R to S and beyond Q to T. The exterior angle $\angle PRS = 140^\circ$ and the exterior angle $\angle PQT = 130^\circ$. Find the interior angle $\angle QPR$.

- (A) 50° (B) 40°
(C) 90° (D) 80°

3. Two sides of a triangle are 18 cm and 7 cm and the third side is a whole number. The third side is then made the smallest it can be, AND separately the largest it can be. What is the SUM of these two extreme third sides?

(A) 36 cm

(B) 35 cm

(C) 37 cm

(D) 25 cm

4. A triangle has sides 9 cm, 12 cm and 15 cm. Classified by BOTH sides and angles, it is best described as:

(A) Scalene and right-angled

(B) Scalene and acute

(C) Isosceles and right-angled

(D) Scalene and obtuse

5. In triangle ABC, $\angle A = (3x + 12)^\circ$, $\angle B = (2x - 6)^\circ$ and $\angle C = (4x - 6)^\circ$. Find the measure of the LARGEST angle.

(A) 72°

(B) 74°

(C) 34°

(D) 70°

6. A triangle has two angles measuring 38° and 109° . Its three sides are listed from shortest to longest. Which ordering is correct?

(A) side opposite $33^\circ <$ side opposite $38^\circ <$ side opposite 109°

(B) side opposite $109^\circ <$ side opposite $38^\circ <$ side opposite 33°

(C) side opposite $38^\circ <$ side opposite $33^\circ <$ side opposite 109°

(D) all three sides are equal

7. In triangle ABC, side BC is extended beyond C to D. The exterior angle $\angle ACD = 5\angle A$ and the interior $\angle B = 24^\circ$. Find $\angle A$.

(A) 30°

(B) 12°

(C) 6°

(D) 24°

8. Two sides of a triangle are 20 cm and 13 cm. How many whole-number third sides make the triangle isosceles (at least two equal sides)?

(A) 2

(B) 1

(C) 3

(D) 0

9. The three exterior angles of triangle ABC (one at each vertex) are $(7y)^\circ$ at A, $(5y)^\circ$ at B and $(6y)^\circ$ at C. Find the interior angle at C.

(A) 120°

(B) 60°

(C) 40°

(D) 80°

10. An isosceles triangle has its two equal sides each 8 cm and a base b that is a whole number. What is the smallest whole-number base b for which the apex angle (the angle opposite the base) is obtuse?

(A) 11 cm

(B) 8 cm

(C) 16 cm

(D) 12 cm

11. In triangle ABC, $\angle A = 90^\circ$. The altitude from A meets the hypotenuse BC at H. If $\angle B = 35^\circ$, what is $\angle HAC$?

(A) 55°

(B) 35°

(C) 45°

(D) 90°

12. In triangle ABC, the bisectors of $\angle B$ and $\angle C$ meet at I, and it is found that $\angle BIC = 116^\circ$. Find $\angle A$.

- (A) 52° (B) 64°
(C) 58° (D) 32°

13. Two sides of a triangle are 21 cm and 8 cm. The third side is a whole number that is a multiple of 3. How many such values are possible?

- (A) 4 (B) 5
(C) 6 (D) 9

14. In triangle ABC, $\angle A - \angle B = 30^\circ$ and $\angle A - \angle C = 54^\circ$. Find the largest angle.

- (A) 88° (B) 58°
(C) 34° (D) 84°

15. In a right-angled triangle the right angle is at C. The altitude from C to the hypotenuse AB has foot H. Which statement is GUARANTEED true for every such triangle?

- (A) H lies strictly between A and B, and CH is perpendicular to AB
(B) H coincides with the midpoint of AB (C) CH equals half of AB
(D) H lies outside segment AB

16. In triangle ABC, $\angle A : \angle B : \angle C = 3 : 4 : 5$ and the sides opposite them are a, b, c. List the sides from longest to shortest.

- (A) $a > b > c$ (B) $b > c > a$
(C) $a > c > b$ (D) $c > b > a$

17. A triangle has sides 5 cm, 6 cm and 8 cm. Classified by its angles using the longest-side test, it is:

- (A) Right-angled (B) Acute
(C) Obtuse (D) Equilateral

18. In triangle ABC, side AB is extended beyond B to E and side AC is extended beyond C to F. The exterior angle $\angle CBE = 112^\circ$ and the exterior angle $\angle BCF = 124^\circ$. Find $\angle A$.

- (A) 124° (B) 56°
(C) 68° (D) 64°

19. Two sides of a triangle are 30 cm and 11 cm. The third side is a whole number. By how much does the count of valid EVEN third sides exceed the count of valid ODD third sides?

- (A) 0 (B) 1
(C) 2 (D) 21

20. In triangle ABC, $AB = AC$ and the median from A meets BC at M. It is given that $\angle BAM = 27^\circ$. What is the measure of $\angle B$?

- (A) 63° (B) 27°
(C) 54° (D) 126°

21. In triangle ABC, $\angle A = (2x + 15)^\circ$, $\angle B = (x + 20)^\circ$ and $\angle C = (3x + 25)^\circ$. Classified by its angles, the triangle is:

- (A) Right-angled
- (B) Obtuse
- (C) Equiangular
- (D) Acute

22. A triangle has perimeter 60 cm and its sides are in the ratio 5 : 12 : 13. By its angles the triangle is, and its longest side is:

- (A) Acute; 26 cm
- (B) Right-angled; 20 cm
- (C) Right-angled; 26 cm
- (D) Obtuse; 26 cm

23. Four candidate side-triples (in cm) are: (9, 9, 17), (6, 7, 13), (8, 15, 17), (5, 5, 11). How many of these form a valid triangle?

- (A) 1
- (B) 3
- (C) 4
- (D) 2

24. In triangle ABC, side BC is extended beyond C to D. The bisector of the exterior angle $\angle ACD$ is drawn. If $\angle A = 64^\circ$ and $\angle B = 48^\circ$, what does each half of the bisected exterior angle measure?

- (A) 112°
- (B) 56°
- (C) 28°
- (D) 44°

25. In an isosceles triangle ABC with $AB = AC$, a point D is taken on BC so that $AD = BD$. If $\angle A = 80^\circ$, find $\angle DAC$.

- (A) 50°
- (B) 30°
- (C) 20°
- (D) 80°

26. Two sides of a triangle are 14 cm and 14 cm. The third side is a whole number making the triangle ACUTE-angled. What is the largest such whole-number third side?

- (A) 20 cm
- (B) 27 cm
- (C) 18 cm
- (D) 19 cm

27. In triangle ABC, $\angle A = 3\angle C$ and $\angle B = \angle A + \angle C - 20^\circ$. Find $\angle B$.

- (A) 100°
- (B) 75°
- (C) 80°
- (D) 60°

28. A triangle has sides 7 cm, 8 cm and 12 cm. A second triangle has angles 50° , 60° and 70° . Which single statement classifies BOTH correctly by angles?

- (A) The first is obtuse; the second is acute
- (B) The first is right-angled; the second is acute
- (C) The first is obtuse; the second is right-angled
- (D) Both are acute

29. In triangle ABC, $\angle B = \angle C$. The bisector of $\angle B$ meets AC at D, and it turns out $\angle BDC = 105^\circ$. Find the vertex angle $\angle A$.

- (A) 50° (B) 30°
(C) 80° (D) 70°

30. A triangle has all three sides different, and its longest side equals 13 while the other two are 5 and 12. The point where its three altitudes would meet (the orthocentre) lies:

- (A) at the midpoint of the longest side (B) outside the triangle
(C) at the right-angle vertex (D) at the centroid

31. Two sides of a triangle are 16 cm and 9 cm, and the perimeter must be a whole number greater than 40 cm. What is the smallest whole-number perimeter satisfying both the triangle inequality and this condition?

- (A) 41 cm (B) 40 cm
(C) 42 cm (D) 26 cm

32. In triangle ABC, $\angle A = 100^\circ$ and $\angle B = \angle C$. The bisectors of $\angle B$ and $\angle C$ meet at I. Find $\angle BIC$.

- (A) 40° (B) 140°
(C) 80° (D) 130°

33. A triangle has sides that are three consecutive whole numbers and its perimeter is 9 cm. Classified by its angles, the triangle is:

- (A) Acute (B) Right-angled
(C) Obtuse (D) Equilateral

34. In triangle ABC, $\angle A = 2\angle B$ and $\angle C = \angle A + \angle B + 30^\circ$. Find the largest angle.

- (A) 75° (B) 105°
(C) 90° (D) 120°

35. A triangle has angles 20° , 20° and 140° . Two altitudes are dropped from the two 20° vertices. Where do the feet of these two altitudes lie?

- (A) inside the triangle, between the endpoints of the opposite sides
(B) exactly at the 140° vertex (C) at the midpoints of the opposite sides
(D) outside the triangle, on the extensions of the opposite sides

36. In triangle ABC, $\angle A = 36^\circ$. Points D on AB and E on AC are chosen so that $AD = AE$, and then DE is joined. Inside triangle ADE, the line from A is NOT drawn; instead, what is the measure of the base angle $\angle ADE$?

- (A) 72° (B) 36°
(C) 108° (D) 54°

37. Two sides of a triangle are 19 cm and 6 cm. The third side is a whole number. The product of the smallest possible and the largest possible third side is:

- (A) 322 (B) 300
(C) 336 (D) 350

38. In triangle ABC, the exterior angle at A equals 3 times the interior angle at A, and $\angle B = 2\angle C$. Find $\angle C$.

- (A) 45° (B) 90°
(C) 30° (D) 60°

39. A triangle has angles 30° , 30° and 120° . By BOTH sides and angles it is best described as:

- (A) Scalene and obtuse (B) Isosceles and acute
(C) Equilateral and obtuse (D) Isosceles and obtuse

40. In triangle ABC, $\angle A = 90^\circ$ and $\angle B = 30^\circ$. The bisector of $\angle A$ meets BC at D. Find $\angle ADB$.

- (A) 105° (B) 75°
(C) 60° (D) 45°

41. A triangle has two sides 25 cm and 25 cm and a base that is a whole number. For how many whole-number bases is the triangle ACUTE-angled?

- (A) 34 (B) 36
(C) 35 (D) 49

42. In triangle ABC, $\angle B = 70^\circ$, $\angle C = 60^\circ$, and AD is the bisector of $\angle A$ meeting BC at D. Find $\angle ADC$.

- (A) 85° (B) 60°
(C) 25° (D) 95°

43. In triangle ABC, the angles satisfy $\angle A = \angle B + \angle C - 40^\circ$. Find $\angle A$.

- (A) 110° (B) 40°
(C) 70° (D) 90°

44. A triangle has sides 11 cm and 13 cm and a third side that is a whole number making it isosceles. How many of these valid isosceles third-side values give a triangle that is NOT obtuse at any vertex (i.e. acute or right)?

- (A) 1 (B) 2
(C) 0 (D) 3

45. In triangle ABC, $\angle A = 2\angle B = 6\angle C$. Find the largest angle.

- (A) 90° (B) 54°
(C) 120° (D) 108°

46. Four candidate angle-triples are: $(44^\circ, 46^\circ, 90^\circ)$, $(60^\circ, 60^\circ, 60^\circ)$, $(80^\circ, 80^\circ, 20^\circ)$, $(100^\circ, 39^\circ, 41^\circ)$. How many of them describe an ACUTE triangle?

- (A) 1 (B) 3
(C) 4 (D) 2

47. In triangle ABC, $AB = AC$ and $\angle A = 20^\circ$. The bisector of base angle B meets AC at D. Find $\angle BDA$.

- (A) 120° (B) 100°
(C) 80° (D) 40°

48. Two sides of a triangle are 50 cm and 11 cm. The third side is a whole number. How many whole-number values can the third side take?

- (A) 21 (B) 22
(C) 20 (D) 39

49. A triangle has angles in the ratio 1 : 2 : 3, and its smallest side is 7 cm. Which side is the longest, and what kind of triangle is it by angles?

- (A) the smallest side stays longest; acute (B) all sides equal; equiangular
(C) the middle side is longest; obtuse
(D) the side opposite the largest angle is longest; right-angled

50. In triangle ABC, $\angle A = 50^\circ$ and $\angle B = 70^\circ$. Side BC is extended beyond C to D. Which is larger, the interior angle $\angle ACB$ or the exterior angle $\angle ACD$, and by how much?

- (A) The interior $\angle ACB$ is larger by 60° (B) The exterior $\angle ACD$ is larger by 60°
(C) They are equal (D) The exterior $\angle ACD$ is larger by 120°

51. A triangle has sides 10 cm, 24 cm and 26 cm. From the vertex opposite the 26 cm side, an altitude is dropped to the 26 cm side. Which statement is correct?

- (A) The triangle is obtuse, so the foot of this altitude falls outside the 26 cm side
(B) The triangle is isosceles, so the foot is the midpoint of the 26 cm side
(C) The triangle is right-angled at that vertex, so the altitude is a genuine perpendicular falling strictly inside the triangle
(D) No such altitude exists because $10 + 24$ equals 26, failing the triangle inequality

52. In triangle ABC, $\angle A = 84^\circ$ and the bisectors of $\angle B$ and $\angle C$ meet at I. The bisectors of $\angle B$ and $\angle C$ are then extended to meet the opposite sides; but for $\angle BIC$ use the incentre. By how many degrees does $\angle BIC$ exceed $\angle A$?

- (A) 6° (B) 90°
(C) 42° (D) 48°

53. A triangle has two sides 12 cm and 16 cm. The third side is a whole number making the triangle right-angled with the third side as the HYPOTENUSE. The third side is:

- (A) 28 cm (B) 4 cm
(C) 20 cm (D) 11 cm

54. In triangle ABC, $\angle A = 50^\circ$, $\angle B = 60^\circ$, $\angle C = 70^\circ$. The bisector of $\angle B$ meets AC at D. Find $\angle BDA$.

- (A) 80° (B) 70°
(C) 110° (D) 100°

55. In triangle ABC, two exterior angles (at A and at B) measure 115° and 125° . Side BC is then extended to form the exterior angle at C. That exterior angle at C measures:

- (A) 240° (B) 60°
(C) 115° (D) 120°

56. A triangle has angles $(x + 5)^\circ$, $(3x - 25)^\circ$ and $(2x + 20)^\circ$. The largest angle measures:

- (A) 65° (B) 35°
(C) 80° (D) 95°

57. In an isosceles triangle ABC with $AB = AC$, the perpendicular from A meets BC at D, and $BD = 9$ cm. A second point E lies on BC with $DE = 4$ cm and E between D and C. What is the length of EC?

- (A) 13 cm (B) 5 cm
(C) 4 cm (D) 9 cm

58. In triangle ABC, $\angle A = 40^\circ$. The bisectors of the EXTERIOR angles at B and C meet at a point J (outside the triangle). The angle $\angle BJC$ equals $90^\circ - \frac{1}{2}\angle A$. Compute $\angle BJC$.

- (A) 110° (B) 20°
(C) 90° (D) 70°

59. Two sides of a triangle are 6 cm and 11 cm. Among all whole-number third sides, exactly one makes the triangle isosceles AND obtuse. That third side is:

- (A) 11 cm (B) 5 cm
(C) 6 cm (D) 16 cm

60. In triangle ABC, $AB = AC$ and side BC is extended beyond C to D. The exterior angle $\angle ACD$ measures 124° . Find the vertex angle $\angle BAC$.

- (A) 56° (B) 68°
(C) 62° (D) 124°

Solutions — Mathematics (Class 7), Chapter 7: A Tale of Three Intersecting Lines — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	B	21	D	31	A	41	C	51	C
2	C	12	A	22	C	32	B	42	D	52	D
3	A	13	B	23	D	33	C	43	C	53	C
4	A	14	A	24	B	34	B	44	B	54	D
5	B	15	A	25	B	35	D	45	D	55	D
6	A	16	D	26	D	36	A	46	D	56	C
7	C	17	C	27	C	37	C	47	A	57	B
8	A	18	B	28	A	38	A	48	A	58	D
9	B	19	B	29	C	39	D	49	D	59	C
10	D	20	A	30	C	40	A	50	B	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) The angles total 180° , split as $2k + 3k + 5k = 180^\circ$, so $k = 18^\circ$; the angles are 36° , 54° and 90° . The largest is 90° , and a bisector halves it into two angles of 45° each. The 90° choice is the whole angle (not halved); 36° and 30° come from halving a wrong angle.

2. (C) The interior angle at R = $180^\circ - 140^\circ = 40^\circ$; the interior angle at Q = $180^\circ - 130^\circ = 50^\circ$. Then $\angle QPR = 180^\circ - 40^\circ - 50^\circ = 90^\circ$. Adding the two interior angles ($40^\circ + 50^\circ = 90^\circ$) and stopping there gives the trap value; 50° and 40° are the other two interiors.

3. (A) The third side x satisfies $18 - 7 < x < 18 + 7$, i.e. $11 < x < 25$. The smallest whole number is 12, the largest is 24, and $12 + 24 = 36$. Using $\leq$ at the ends would wrongly give 11 and 25 (sum 36 too, but for the wrong reason) — the correct strict bounds give 12 and 24. The 25 distractor is the forbidden $18+7$.

4. (A) All three sides differ, so it is scalene. Checking the largest side: $9^2 + 12^2 = 81 + 144 = 225 = 15^2$, so the angle opposite the 15 cm side is exactly 90° — a right triangle. Acute

would need the square of the longest to be LESS than the sum of the other squares; obtuse would need it greater; here it is equal. No two sides are equal, so it is not isosceles.

5. (B) Sum: $(3x+12)+(2x-6)+(4x-6) = 9x = 180$, so $x = 20$. Then $\angle A = 72^\circ$, $\angle B = 34^\circ$, $\angle C = 74^\circ$ (check $72+34+74 = 180$). The largest is $\angle C = 74^\circ$. The 72° distractor is $\angle A$ (the second-largest), 34° is the smallest, and 70° is a slip.

6. (A) The third angle is $180^\circ - 38^\circ - 109^\circ = 33^\circ$. The longest side lies opposite the largest angle (109°) and the shortest opposite the smallest (33°), giving $33^\circ < 38^\circ < 109^\circ$ for the side order shortest to longest. Reversing the chain pairs the longest side with the smallest angle, which is the classic inversion error.

7. (C) The exterior angle equals the sum of the two remote interiors: $\angle ACD = \angle A + \angle B$, so $5\angle A = \angle A + 24^\circ$, giving $4\angle A = 24^\circ$ and $\angle A = 6^\circ$. Treating the exterior as equal to $\angle A$ alone, or as the adjacent interior, breaks the relation; 30° would be $5\angle A$, not $\angle A$.

8. (A) Valid third side x satisfies $7 < x < 33$. For isosceles, the third side must equal one of the given sides: $x = 20$ (valid, since $7 < 20 < 33$) or $x = 13$ (valid, since $7 < 13 < 33$). Both lie in range, giving 2 values. Forgetting that BOTH equalities are allowed gives 1; the answer is not 0 because both 13 and 20 are in range.

9. (B) The three exterior angles of any triangle sum to 360° : $7y + 5y + 6y = 18y = 360$, so $y = 20$. The exterior angle at C is $6y = 120^\circ$, so the interior angle at C is its supplement, $180^\circ - 120^\circ = 60^\circ$. The 120° distractor is the exterior (not interior) angle at C; 40° and 80° are the interiors at A and B.

10. (D) With equal sides 8 and 8, the apex angle (opposite b) is a right angle when $b^2 = 8^2 + 8^2 = 128$, i.e. $b = \sqrt{128} \approx 11.31$, and obtuse when $b^2 > 128$. Since $11^2 = 121 < 128$ but $12^2 = 144 > 128$, the smallest whole-number base giving an obtuse apex is 12. $b = 11$ still gives an acute apex; $b = 16$ is degenerate ($8 + 8 = 16$ fails the inequality).

11. (B) In right triangle ABH, $\angle AHB = 90^\circ$ and $\angle B = 35^\circ$, so $\angle BAH = 55^\circ$. Since $\angle BAC = 90^\circ$, $\angle HAC = 90^\circ - 55^\circ = 35^\circ$. (Equivalently, in triangle AHC, $\angle AHC = 90^\circ$ and $\angle C = 55^\circ$, giving $\angle HAC = 35^\circ$.) The 55° trap is $\angle BAH$, not $\angle HAC$.

12. (A) The angle between the two interior bisectors is $\angle BIC = 90^\circ + \frac{1}{2}\angle A$. So $116^\circ = 90^\circ + \frac{1}{2}\angle A$ gives $\frac{1}{2}\angle A = 26^\circ$, $\angle A = 52^\circ$. Forgetting the $+90^\circ$ and writing $116 = \frac{1}{2}\angle A$ gives 232 (impossible); using $90 - \frac{1}{2}\angle A$ gives a negative; 64° would be $\angle A$ if $\angle BIC$ were 122° .

13. (B) Third side x satisfies $21 - 8 < x < 21 + 8$, i.e. $13 < x < 29$, so x runs over 14 to 28. The multiples of 3 in that range are 15, 18, 21, 24 and 27 — five values. Including 12 (just below the range) or 30 (just above) would give 6 or 7; 4 drops one valid multiple.

14. (A) Let $\angle A = a$. Then $\angle B = a - 30$ and $\angle C = a - 54$. Sum: $3a - 84 = 180$, so $3a = 264$, $a = 88^\circ$. Since both differences are positive, $\angle A$ is the largest at 88° . Then $\angle B = 58^\circ$ and $\angle C =$

34° (check $88+58+34 = 180$). The distractors are the two smaller angles and a sign-slip.

15. (A) In a triangle that is right-angled at C, both base angles A and B are acute, so the foot of the perpendicular from C falls strictly inside AB, and by definition $CH \perp AB$. H is the midpoint of AB only when the triangle is also isosceles, which is not guaranteed; $CH =$ half of AB is a special (isosceles) case; H falls outside only when the foot is dropped from an acute vertex of an OBTUSE triangle, not here.

16. (D) Angles are 45° , 60° , 75° (since $3k+4k+5k=180$, $k=15$). The largest angle $75^\circ = \angle C$, so its opposite side c is longest; the smallest angle $45^\circ = \angle A$, so a is shortest. Hence $c > b > a$. The $a > b > c$ choice inverts the rule (longest side opposite SMALLEST angle), the classic trap.

17. (C) Compare the square of the longest side ($8^2 = 64$) with the sum of the squares of the other two ($5^2 + 6^2 = 25 + 36 = 61$). Since $64 > 61$, the angle opposite the longest side is obtuse, so the triangle is obtuse. Right would need $64 = 61$; acute would need $64 < 61$. The sides are all different, so not equilateral.

18. (B) $\angle CBE$ is the exterior angle at B (equal to $\angle A + \angle C$) and $\angle BCF$ is the exterior angle at C (equal to $\angle A + \angle B$). Adding: $(\angle A + \angle C) + (\angle A + \angle B) = 112 + 124 = 236$, i.e. $2\angle A + (\angle B + \angle C) = 236$. Since $\angle B + \angle C = 180 - \angle A$, we get $2\angle A + 180 - \angle A = 236$, so $\angle A + 180 = 236$, $\angle A = 56^\circ$. The 124° distractor is one exterior angle; 68° and 64° are the interior B and C.

19. (B) Third side x satisfies $30 - 11 < x < 30 + 11$, i.e. $19 < x < 41$, so x runs over the whole numbers 20 through 40 — that is 21 values in all. Among 20..40 the even values are 20, 22, ..., 40 (11 of them) and the odd values are 21, 23, ..., 39 (10 of them). Evens exceed odds by $11 - 10 = 1$. The 21 distractor is the total count; 0 wrongly assumes equal parity counts on a 21-long run.

20. (A) In an isosceles triangle with $AB = AC$, the median from the apex A is also the altitude, so $\angle AMB = 90^\circ$. In triangle ABM, $\angle BAM + \angle B + \angle AMB = 180^\circ$, giving $27^\circ + \angle B + 90^\circ = 180^\circ$, so $\angle B = 63^\circ$. The 27° distractor confuses $\angle B$ with $\angle BAM$; 54° is the full apex angle $2 \times 27^\circ$.

21. (D) Sum: $(2x+15)+(x+20)+(3x+25) = 6x + 60 = 180$, so $x = 20$. Then $\angle A = 55^\circ$, $\angle B = 40^\circ$, $\angle C = 85^\circ$. All three are below 90° , so the triangle is acute. The largest, 85° , is close to but not 90° , so it is not right; none exceeds 90° , so not obtuse.

22. (C) $5k+12k+13k = 30k = 60$, so $k = 2$; sides are 10, 24, 26 cm. Test the longest: $10^2 + 24^2 = 100 + 576 = 676 = 26^2$, so the triangle is right-angled with longest side 26 cm. Acute/obtuse fail the equality test; 20 cm would be $10k$, not the longest.

23. (D) (9,9,17): $9+9=18 > 17$ ✓ valid. (6,7,13): $6+7=13$, not greater than 13, so degenerate ✗. (8,15,17): $8+15=23 > 17$ ✓ valid. (5,5,11): $5+5=10 < 11$ ✗. Two valid. The trap is counting

(6,7,13) as valid because the sum equals the third side, but equality fails the strict inequality.

24. (B) The exterior angle $\angle ACD = \angle A + \angle B = 64^\circ + 48^\circ = 112^\circ$. Its bisector splits it into two equal parts of $112^\circ \div 2 = 56^\circ$ each. The 112° distractor is the whole exterior angle (not halved); 28° halves a wrong base; 44° is unrelated.

25. (B) Since $AB = AC$, the base angles are equal: $\angle B = \angle C = (180 - 80)/2 = 50^\circ$. In triangle ABD , $AD = BD$ makes it isosceles with $\angle DAB = \angle B = 50^\circ$. Then $\angle DAC = \angle BAC - \angle DAB = 80^\circ - 50^\circ = 30^\circ$. The 50° distractor is $\angle DAB$ itself; 20° and 80° come from misassigning the equal angles.

26. (D) With equal sides 14 and 14, the angle opposite the third side b is acute when $b^2 < 14^2 + 14^2 = 392$, i.e. $b < \sqrt{392} \approx 19.8$. The two base angles are always acute, so the triangle is acute exactly when $b^2 < 392$, giving $b \leq 19$ ($19^2 = 361 < 392$ but $20^2 = 400 > 392$). The largest whole b is 19. $b = 20$ makes the apex obtuse; 27 is near the degenerate bound.

27. (C) Let $\angle C = c$. Then $\angle A = 3c$ and $\angle B = 3c + c - 20 = 4c - 20$. Sum: $3c + (4c - 20) + c = 8c - 20 = 180$, so $8c = 200$, $c = 25^\circ$. Then $\angle A = 75^\circ$, $\angle C = 25^\circ$, $\angle B = 4(25) - 20 = 80^\circ$ (check $75+80+25 = 180$). The 75° distractor is $\angle A$, not $\angle B$; 100° drops the -20° .

28. (A) First triangle: longest side 12, and $7^2 + 8^2 = 49 + 64 = 113 < 144 = 12^2$, so the angle opposite 12 is obtuse $\rightarrow$ obtuse triangle. Second triangle: all of $50^\circ, 60^\circ, 70^\circ$ are below $90^\circ \rightarrow$ acute. So the first is obtuse and the second is acute. The first is not right ($113 \neq 144$); the second has no 90° angle.

29. (C) Let $\angle B = \angle C = b$, so $\angle A = 180 - 2b$. BD bisects $\angle B$, so $\angle DBC = b/2$. In triangle BDC the angles $\angle DBC = b/2$, $\angle C = b$ and $\angle BDC = 105^\circ$ sum to 180° : $b/2 + b + 105 = 180$, giving $3b/2 = 75$, so $b = 50^\circ$. Then $\angle A = 180 - 2(50) = 80^\circ$. The 50° distractor is each base angle b ; 30° is the half-angle remainder; 70° is a slip.

30. (C) Since $5^2 + 12^2 = 25 + 144 = 169 = 13^2$, the triangle is right-angled at the vertex opposite the 13-side. In a right triangle, two of the altitudes are the legs themselves and all three altitudes pass through the right-angle vertex, so the orthocentre is exactly that vertex. The orthocentre lies outside only for an obtuse triangle; the midpoint of the hypotenuse is the circumcentre, not the orthocentre.

31. (A) The third side x satisfies $7 < x < 25$, so the perimeter $25 + x$ lies in $32 < \text{perimeter} < 50$. We also need perimeter > 40 and whole. The smallest whole number above 40 within (32,50) is 41 (corresponding to $x = 16$, which is in range). 40 is not greater than 40; 26 ignores the >40 condition.

32. (B) The angle between the two interior bisectors is $\angle BIC = 90^\circ + \frac{1}{2}\angle A = 90^\circ + \frac{1}{2}(100^\circ) = 140^\circ$. (The fact that $\angle B = \angle C$ is extra information not needed for this formula.) The 40° distractor is the base angle $\angle B$ itself; 80° is $\angle B + \angle C$; 130° uses a wrong half-angle.

33. (C) Three consecutive numbers summing to 9 are 2, 3, 4 (and $2+3 = 5 > 4$, so it is a valid triangle). Test the longest: $2^2 + 3^2 = 4 + 9 = 13$, while $4^2 = 16$. Since $16 > 13$, the angle opposite the 4 cm side is obtuse, so the triangle is obtuse. Acute would need $16 < 13$; right would need $16 = 13$; the sides are unequal so not equilateral.

34. (B) Let $\angle B = b$, $\angle A = 2b$, $\angle C = 2b + b + 30 = 3b + 30$. Sum: $2b + b + (3b + 30) = 6b + 30 = 180$, so $6b = 150$, $b = 25^\circ$. Then $\angle A = 50^\circ$, $\angle B = 25^\circ$, $\angle C = 105^\circ$ (check $50+25+105 = 180$). The largest is $\angle C = 105^\circ$. The 75° distractor is $\angle A + \angle B$; 90° and 120° come from arithmetic slips.

35. (D) The triangle is obtuse ($140^\circ > 90^\circ$). When an altitude is dropped from a vertex of an acute angle in an obtuse triangle, its foot lands on the EXTENSION of the opposite side, i.e. outside the triangle. Only the altitude from the obtuse vertex lands inside. Feet at midpoints would be for medians; the feet do not coincide with the obtuse vertex.

36. (A) Triangle ADE has $AD = AE$, so it is isosceles with apex angle $\angle A = 36^\circ$. The two base angles are equal and sum with 36° to 180° : each $= (180 - 36)/2 = 72^\circ$. So $\angle ADE = 72^\circ$. The 36° distractor is the apex; 108° is the supplement of a base angle; 54° halves wrongly.

37. (C) The third side x satisfies $13 < x < 25$, so smallest $= 14$, largest $= 24$. Their product is $14 \times 24 = 336$. Using 13×25 (the forbidden equal-sum endpoints) gives 325; $322 = 14 \times 23$ (wrong largest); $350 = 14 \times 25$ mixes one wrong endpoint.

38. (A) The exterior angle at A is $180^\circ - \angle A$, and it equals $3\angle A$, so $180 - \angle A = 3\angle A$, giving $4\angle A = 180$, $\angle A = 45^\circ$. Then $\angle B + \angle C = 135^\circ$, and with $\angle B = 2\angle C$ we get $3\angle C = 135^\circ$, so $\angle C = 45^\circ$ (and $\angle B = 90^\circ$). The 90° distractor is $\angle B$; 30° and 60° come from misusing the exterior relation.

39. (D) Two angles are equal (30° and 30°), so the sides opposite them are equal — the triangle is isosceles. One angle is $120^\circ > 90^\circ$, so it is obtuse. It cannot be scalene (two equal angles force two equal sides), nor acute (120° exceeds 90°), nor equilateral (the angles are not all 60°).

40. (A) $\angle C = 180 - 90 - 30 = 60^\circ$. The bisector of the right angle at A gives $\angle BAD = 45^\circ$. In triangle ABD, $\angle ADB = 180 - \angle B - \angle BAD = 180 - 30 - 45 = 105^\circ$. The 75° distractor is the supplement $\angle ADC$; 45° is $\angle BAD$; 60° is $\angle C$.

41. (C) With equal sides 25, the base b satisfies $0 < b < 50$ for a triangle. The two base angles are always acute, so the triangle is acute exactly when the apex angle (opposite b) is acute, i.e. $b^2 < 25^2 + 25^2 = 1250$, so $b < \sqrt{1250} \approx 35.36$, giving $b \leq 35$. Combined with $b \geq 1$, the whole numbers 1 through 35 work — 35 values. $b = 36$ makes the apex obtuse; 49 counts up to the degenerate bound.

42. (D) $\angle A = 180 - 70 - 60 = 50^\circ$, so its bisector gives $\angle DAC = 25^\circ$. In triangle ADC, $\angle ADC = 180 - \angle DAC - \angle C = 180 - 25 - 60 = 95^\circ$. The 85° distractor is the supplement $\angle ADB$; 25° is $\angle DAC$; 60° is $\angle C$.

43. (C) Since $\angle A + \angle B + \angle C = 180^\circ$, we have $\angle B + \angle C = 180 - \angle A$. Substituting into the given relation: $\angle A = (180 - \angle A) - 40 = 140 - \angle A$, so $2\angle A = 140$, $\angle A = 70^\circ$. The 110° distractor is $\angle B + \angle C$; 40° and 90° are arithmetic traps.

44. (B) Isosceles third sides are 11 (sides 11, 11, 13) or 13 (sides 11, 13, 13); both satisfy $2 < x < 24$, so both are valid. Case 11, 11, 13: longest 13, and $11^2 + 11^2 = 242 > 169 = 13^2 \rightarrow$ acute. Case 11, 13, 13: longest 13, and $11^2 + 13^2 = 290 > 169 \rightarrow$ acute. Both are acute, hence not obtuse — count is 2. The trap is assuming one of them must be obtuse.

45. (D) Let $\angle A = 2\angle B = 6\angle C = k$. Then $\angle A = k$, $\angle B = k/2$, $\angle C = k/6$. Sum: $k + k/2 + k/6 = (6k + 3k + k)/6 = 10k/6 = 5k/3 = 180$, so $k = 108^\circ$. The largest is $\angle A = 108^\circ$. Then $\angle B = 54^\circ$, $\angle C = 18^\circ$ (check $108 + 54 + 18 = 180$). The 54° distractor is $\angle B$; 90° and 120° are slips.

46. (D) Acute means ALL three angles below 90° . (44,46,90): has a 90° , right not acute. (60,60,60): all below $90^\circ \rightarrow$ acute $\checkmark$. (80,80,20): all below $90^\circ \rightarrow$ acute $\checkmark$. (100,39,41): $100^\circ > 90^\circ \rightarrow$ obtuse. So exactly 2 are acute. The trap is counting the 90° triple as acute.

47. (A) Base angles: $\angle B = \angle C = (180 - 20)/2 = 80^\circ$. BD bisects $\angle B$, so $\angle ABD = 40^\circ$. In triangle ABD, $\angle A = 20^\circ$ and $\angle ABD = 40^\circ$, so $\angle BDA = 180 - 20 - 40 = 120^\circ$. The 100° distractor is the supplement $\angle BDC$; 80° is the full base angle; 40° is $\angle ABD$.

48. (A) The third side x satisfies $50 - 11 < x < 50 + 11$, i.e. $39 < x < 61$. The whole numbers are 40 through 60 inclusive, a count of $60 - 40 + 1 = 21$. Using $\leq$ at the ends would wrongly add 39 and 61; 39 is the difference (a tempting but wrong endpoint count).

49. (D) Angles $1k + 2k + 3k = 180$ give $k = 30^\circ$, so $30^\circ, 60^\circ, 90^\circ$ — a right triangle. The longest side lies opposite the 90° angle (the largest), not opposite the smallest. The 7 cm smallest side faces the 30° angle. So it is right-angled with the longest side opposite the right angle.

50. (B) The interior angle $\angle ACB = 180^\circ - 50^\circ - 70^\circ = 60^\circ$. The exterior angle $\angle ACD = \angle A + \angle B = 50^\circ + 70^\circ = 120^\circ$ (also its supplement $180^\circ - 60^\circ = 120^\circ$). So the exterior exceeds the interior by $120^\circ - 60^\circ = 60^\circ$. Treating the exterior as equal to the interior, or reversing which is larger, are the traps; 120° is the exterior angle, not the difference.

51. (C) Since $10^2 + 24^2 = 100 + 576 = 676 = 26^2$, the triangle is right-angled at the vertex opposite the 26 cm side (the hypotenuse). From that right-angle vertex the altitude to the hypotenuse is a true perpendicular landing strictly inside the triangle. It is not obtuse (the square test gives equality, not greater-than), the sides are unequal so it is not isosceles, and $10 + 24 = 34 > 26$ so the inequality holds.

52. (D) $\angle BIC = 90^\circ + \frac{1}{2}\angle A = 90^\circ + 42^\circ = 132^\circ$. The excess over $\angle A$ is $132^\circ - 84^\circ = 48^\circ$. The 6° distractor is $90 - \frac{1}{2}\angle A$; 42° is $\frac{1}{2}\angle A$; 90° forgets to subtract $\angle A$.

53. (C) If the third side c is the hypotenuse (longest), then $c^2 = 12^2 + 16^2 = 144 + 256 = 400$, so $c = 20$ cm, which satisfies $4 < c < 28$. The 28 distractor is the forbidden sum; 4 is the difference; 11 would be a leg case ($\sqrt{16^2 - 12^2}$ is not whole here).

54. (D) The bisector of $\angle B$ gives $\angle ABD = 30^\circ$. In triangle ABD , $\angle A = 50^\circ$, $\angle ABD = 30^\circ$, so $\angle BDA = 180 - 50 - 30 = 100^\circ$. The 80° distractor is its supplement $\angle BDC$; 70° is $\angle C$; 110° is a slip.

55. (D) The three exterior angles of any triangle sum to 360° . So exterior at $C = 360 - 115 - 125 = 120^\circ$. (Equivalently, interiors at A and B are 65° and 55° , so interior $C = 60^\circ$ and its exterior is 120° .) The 240° distractor is $115 + 125$; 60° is the interior angle at C , not its exterior.

56. (C) Sum: $(x+5) + (3x-25) + (2x+20) = 6x = 180$, so $x = 30$. The angles are 35° , 65° and 80° (check $35+65+80 = 180$). The largest is 80° . The 65° distractor is the middle angle, 35° the smallest, and 95° a slip.

57. (B) In an isosceles triangle with $AB = AC$, the perpendicular from the apex A bisects the base, so $DC = BD = 9$ cm. With E between D and C and $DE = 4$ cm, $EC = DC - DE = 9 - 4 = 5$ cm. The 13 distractor adds instead of subtracts; 9 cm is DC alone; 4 cm is DE .

58. (D) Using the given external-bisector relation $\angle BJC = 90^\circ - \frac{1}{2}\angle A = 90^\circ - 20^\circ = 70^\circ$. (Contrast with the INTERNAL bisectors, which give $90^\circ + \frac{1}{2}\angle A = 110^\circ$.) The 110° distractor is exactly that internal-bisector value, the classic confusion; 20° is $\frac{1}{2}\angle A$; 90° forgets the subtraction.

59. (C) The valid third side satisfies $5 < x < 17$. For isosceles the third side must equal 6 or 11; both 6 and 11 are in range. Case $x = 6$ (sides 6, 6, 11): longest is 11, and $6^2 + 6^2 = 72 < 121 = 11^2$, so the angle opposite 11 is obtuse — isosceles AND obtuse. Case $x = 11$ (sides 6, 11, 11): longest 11, and $6^2 + 11^2 = 157 > 121$, so it is acute. Hence only $x = 6$ works. The 5 distractor is the forbidden lower bound; 16 is acute.

60. (B) The exterior angle $\angle ACD$ is the supplement of the interior base angle $\angle ACB$, so $\angle ACB = 180^\circ - 124^\circ = 56^\circ$. Since $AB = AC$, the base angles are equal, so $\angle ABC = \angle ACB = 56^\circ$. Then the vertex angle $\angle BAC = 180^\circ - 56^\circ - 56^\circ = 68^\circ$. The 56° distractor is each base angle; 62° halves the exterior; 124° is the exterior angle itself.